WINA PRASETYO

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PERSONAL STATEMENT

Ever since I assembled toy robots in primary school, I've been hooked on robotics and became very fascinated by how engineering can bring machines to life. Now, as a 3rd year Mechatronics student at the University of Canterbury, I'm diving deep into mechanical design, electronics, and programming with the goal of developing smart, efficient robots in the future. I am eager to apply my skills through internships or collaborative projects and connect with others passionate about automation and innovation.

SKILLS

- **Soft Skills:** Problem Solving, Time Management, Adaptability
- **Coding:** C, Python, PLC Programming, JavaScript, HTML/CSS, SQL, VHDL, MATLAB, Simscape
- **Engineering:** Soldering, Welding, Lathe, Milling, CNC Machining, Engineering Drawing, 3D CAD, PCB Design

PROJECTS

https://winap.github.io/Projects/

Line Following Robot

- Designed, manufactured, and tested a custom PCB for a line following robot using KiCad for schematic and PCB layout, supported by online circuit simulation tools. Soldered the board together neatly.
- Programmed and debugged in C using Arduino. Implementing different control algorithms for its navigation, sensor processing and motor control.
- Optimised the mechanical design through iterative prototyping, testing, and troubleshooting using 3D CAD.

Step Counter

- Developed an embedded step counter system using a microcontroller-based board and C programming.
- Implemented signal processing techniques, interrupt-driven algorithms, kernel-driven programming concepts, and event handling for accurate step detection.
- Applied embedded software development, sensor interfacing, debugging, and real-time system design skills.

Solar Car

- Investigated the characteristics of the solar panel and DC motor to determine an appropriate design strategy.
- Assisted with the design, simulation, assembly, and testing of a buck converter PCB to interface the solar panel with the DC motor using LTspice and KiCad.
- Integrated the electrical power system into the vehicle and conducted iterative performance testing to optimise efficiency and reliability under solar power.

Tap Wrench Project

- Manufactured a functional tap wrench from raw steel stock using workshop machining processes.
- Applied milling, turning (lathe), drilling, and precision measurement techniques while working from engineering drawings and manufacturing tolerances.
- Developed practical skills in machining, CAD interpretation, manufacturing processes, and mechanical design.

Image Processing Project

- Applied matrix operations and linear transformations to process and enhance digital images using Python.
- Implemented image reconstruction, transformation, and refinement algorithms using mathematical modelling and matrix-based techniques.
- Developed programming skills in computational methods, data processing, and image analysis.

Disrupt Challenge Hackathon

- Collaborated in the design and prototyping of a wearable safety device for remote workers.
- Developed concepts including fall detection, heart-rate monitoring, emergency communication, worker dashboards, and proximity alerts.
- Contributed to hardware/software integration, embedded system design, rapid prototyping, and final presentation.
- Applied skills in electronics design, embedded programming, product development, and teamwork.

EDUCATION

University of Canterbury
2024 – Present
Bachelor of Engineering with Honours in Mechatronics Engineering

- UCIEE Mentor
- Women in Tech (WitSoc) Mentor
- UC CollectiveUni Secretary
- Indonesian Organisation (PPIC) General Exec

Hutt Valley High School
2018 - 2023
NCEA Level 1, 2, & 3
Excellence Endorsed

- SNAP Mentor
- Club Leader
- Peer Support & Tutor

WORK EXPERIENCE

Tutor
Job Postings 2026 - Present

Tutoring students primarily in Physics and Math, using engaging teaching strategies to simplify complex concepts, promoting student understanding and confidence, and adapting tutoring approaches to accommodate individual learning styles, academic goals, and varying levels of ability.

Event Staff Leader
Lower Hutt Event Center 2023 - 2025

Responsible for large team management for event logistics, organisation, operations, and professional maintenance for corporate, social and cultural events.

AWARDS

- 2025 Deloitte Hackathon People's Choice Award
- 2024 University of Canterbury Go Waitaha Canterbury & Hiranga Scholarship
- 2024 Maintenance of Engineering Society of New Zealand Scholarship
- 2024 Hutt Valley High School David Kaye Scholarship
- 2023 Japanese Speech Competition Winner

VOLUNTEERING

- UC Chaplaincy Volunteer
- UC Student Volunteer Army

INTERESTS & HOBBIES

- **Creative pursuits:** Reading, watching movies, and learning music
- **Sports:** Martial arts, badminton and occasionally biking
- **Games:** Sudoku, chess, and jigsaw puzzles
- **Languages:** Currently self-learning Japanese and Korean
- **Food:** Cooking, baking, and experimenting with recipes and cuisines

REFERENCES

References available upon request.